Evaluating Relation Extraction Strategies for Knowledge Graph Construction

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Project Introduction & Setup

Goal: Identify the most reliable relation extraction (RE) setup for automated Knowledge Graph construction.

- Compare heuristic, ML, and neural RE.
- Evaluate entity encoding impact.

Data & Models:

- **Data:** Synthetic, SemEval-2010, AG News, DBPedia, Yelp.
- **Models:** Rules, OpenIE, TF-IDF + (SVM/LR/RF), SBERT.

Methodology

Pipeline: Parse → Encode (`raw/mask/marker`) → Classify → Score.

Model Families:

- **Rules:** High precision, low recall.
- **Classical ML:** Robust baseline.
- **Neural:** SBERT + Torch head.

Metrics:

- **Quality:** Macro-F1, MCC.
- **Ranking:** PR-AUC, ROC-AUC.
- **Cost:** Runtime ratio.

Experimental Results

SemEval-2010 Task 8 (Top Configurations):

Config	Macro-F1	MCC	Mode
SVM + markers	**0.736**	**0.679**	**markers**
NB + markers	0.720	0.676	markers
LogReg + markers	0.718	0.655	markers

Key Performance Insights:

- **Representations:** Markers (0.714) > Raw (0.684) > Masking (0.661).
- **Neural vs Classical:** Neural leads in AG News (+0.076 F1) but is ×150 slower. Classical dominates SemEval (−0.256 neural drop, ×41 faster).
- **Robustness:** Noise affects performance more than class imbalance.

External Validation Results:

Dataset	Winner	F1	Cost Ratio
ag_news	SBERT	0.849	150.6×
dbpedia_14	SBERT	0.946	27.6×
yelp_pol	TF-IDF+LR	0.825	1.0×
sem_eval	TF-IDF+LR	0.630	1.0×

KG Output Policy & Triage

Selected Model: `TF-IDF + LogReg` (Stability + Throughput).

Thresholds per Relation:

Type	Accept	Review
Entity-Dest	0.84	0.71
Cause-Effect	0.83	0.70
Comp-Whole	0.56	0.48
Prod-Prod	0.51	0.43

Queue Statistics:

- **Accept:** 156 (25%)
- **Review:** 103 (17%)
- **Reject:** 360 (58%)

Visual Dashboard

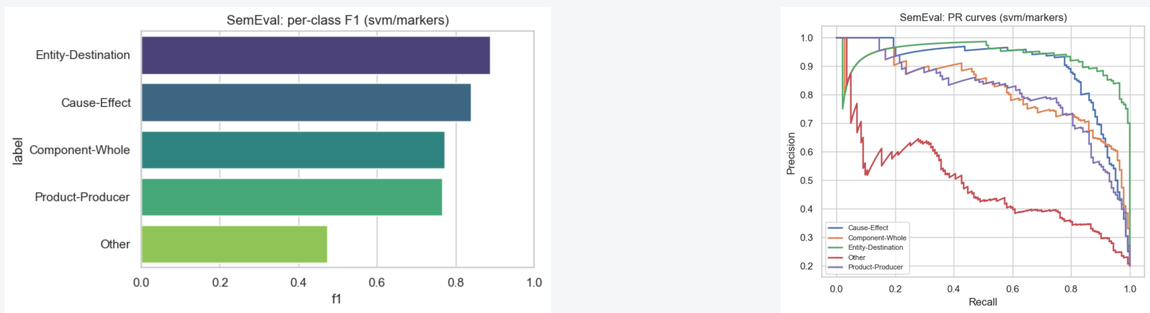

Fig. 1: Data Characteristics

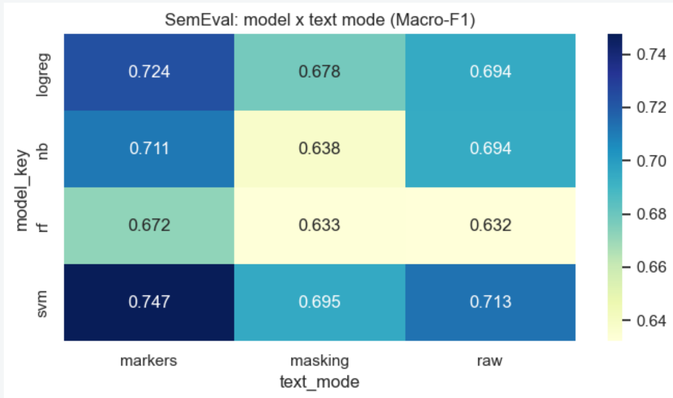

Fig. 2: SemEval Ablation Heatmap

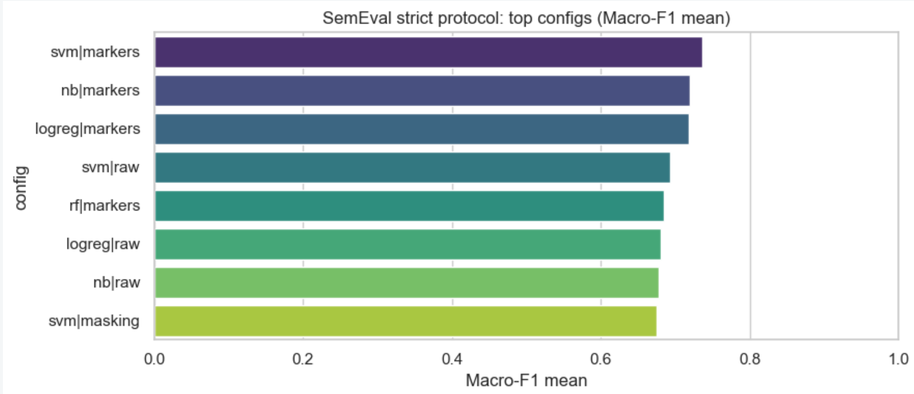

Fig. 3: F1 vs Runtime Efficiency

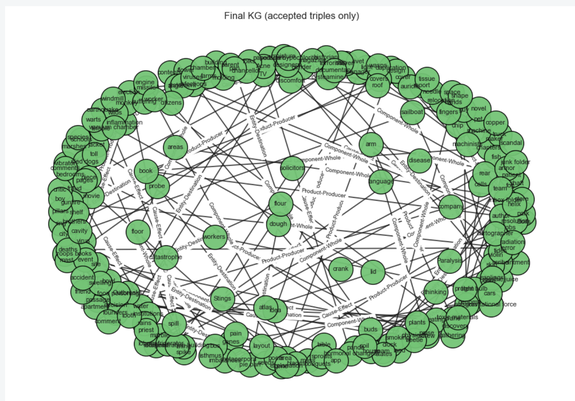

Fig. 4: Final Graph Construction Queueing

Conclusions & Recommendations

- **Entity Representation:** Entity markers are the superior choice for RE, outperforming masking by preserving semantic boundaries.
- **Model Selection:** Classical **SVM/LR** with markers is the optimal production default. It provides the best speed-to-accuracy ratio.
- **Neural Utility:** Neural models (SBERT) should be reserved for high-complexity datasets where quality gain justifies 100× latency.
- **KG Integrity:** A three-tier queue (`accept/review/reject`) based on calibrated thresholds is essential for safe automated graph population.
- **Main Risk:** Relation label confusion is a greater threat to KG quality than extraction failure; thresholding effectively mitigates this risk.

Final Summary

The study confirms that a calibrated policy combining robust classical inference with strict confidence guardrails (**25% auto-accept rate**) ensures a high-fidelity Knowledge Graph while maintaining production-scale throughput.

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